Test 10 – Grade 8

Name: _____
Grade / Class: 8

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"During winter holidays, Omar volunteered at a local library to organize books and read stories to younger children. He noticed how storytelling can inspire creativity and improve language skills. His project also included creating a small reading corner for kids in the neighborhood. By the end of the program, Omar felt motivated to continue community service and promote literacy."

a) Where did Omar volunteer during the holidays?
b) List two activities Omar performed at the library.
c) How did Omar feel after completing the project?
d) Summarize the passage in four sentences using your own words.

2. Grammar & Vocabulary
a) Combine the sentences using a suitable conjunction:
"The sun set behind the hills. The sky turned orange and pink."

b) Identify all nouns and verbs in the sentence:
"The young artists painted murals on the school walls enthusiastically."

c) Correct the grammatical error:
"Each of the students have completed their projects successfully."

d) Rewrite in indirect speech:
"Aisha said, 'I am preparing for the science competition tomorrow.'"

3. Writing
Write a short essay (120–150 words) on the topic: "The Importance of Environmental Conservation". Include examples, challenges, and personal opinion.

Section B: Mathematics

1. Algebra
a) Solve for x:
$7x - 9 = 2x + 16$

b) Factorize:
$x^2 - 11x + 24$

c) Simplify:
$6m^2n - 4mn^2 + 5m^2n$

2. Geometry & Mensuration
a) A parallelogram has a base of 14 cm and height of 9 cm. Find its area.
b) A sphere has a radius of 6 cm. Find its volume.

3. Ratio, Proportion & Percentage
a) A laptop costs $800. It is sold at a 10% discount. Find the selling price.
b) If 15 pencils cost $45, find the cost of 50 pencils.

4. Data Handling
Marks of 10 students in a geography test: 17, 19, 18, 20, 16, 18, 19, 17, 20, 16.
a) Find the mean, median, and mode.
b) Represent the data using a frequency table.

Section C: Science

1. Biology
a) Describe the functions of the human liver.
b) Explain transpiration in plants and its importance.
c) List two differences between reptiles and amphibians.

2. Physics
a) A train accelerates from 10 m/s to 30 m/s in 5 seconds. Calculate its acceleration.
b) Define potential energy and give an example.
c) Explain the difference between reflection and absorption of light.

3. Chemistry
a) Identify whether the following are elements, compounds, or mixtures:

- **Nitrogen**
- **Sugar ($C_{12}H_{22}O_{11}$)**
- **Seawater**

b) Write a balanced chemical equation for the reaction of zinc with sulfuric acid.

4. Environmental Science
a) Name two major greenhouse gases and their effects.
b) Suggest three ways to reduce plastic usage in daily life.

Section D: Social Studies

1. History
a) Name two important events of the Industrial Revolution and their impact.
b) Explain the significance of the Enlightenment in Europe.
c) Identify the main countries in the Allied Powers of World War I.

2. Geography
a) Define ecosystem and give an example.
b) Name the longest river in South America.
c) Explain two environmental effects of mining activities.

3. Civics / Political Science
a) Compare parliamentary and presidential forms of government.
b) State two duties of citizens in a democratic society.

4. Economics
a) Explain the concept of opportunity cost with an example.
b) Differentiate between gross and net profit in business.

Section E: ICT / Computer Studies

1. Computer Basics
a) What is cloud computing? Give one example.
b) Name one input device and one output device.

2. Applications
a) Which spreadsheet function calculates the average value in a dataset?
b) List two uses of graphic design software in schools.

3. Programming Basics
a) Define an array and give an example in Python.
b) Write a Python code snippet to print the first 10 even numbers.

4. Cyber Awareness
a) Explain phishing and one way to prevent it.
b) List two methods to secure personal accounts online.

Section F: Islamiyat

1. Quran & Hadith
a) Mention two ethical teachings from the Quran.
b) Narrate a Hadith emphasizing truthfulness.

2. Aqidah / Belief
a) List the six articles of Iman (faith).
b) Explain the importance of believing in angels in Islam.

3. Fiqh / Worship
a) Explain the significance of Zakah in Islam.
b) Describe the spiritual benefits of Hajj.

4. Islamic Ethics & Morality
a) How does Islam promote kindness to neighbors?
b) Explain the importance of protecting nature according to Islam.

Section G: Arts & Music

1. Visual Arts
a) Explain the concept of symmetry in drawing.
b) Name two tertiary colors.

2. Music
a) Define rhythm and tempo in music.
b) Give an example of a percussion instrument from any culture.

Section H: Health & Physical Education

1. Physical Fitness
a) Name two exercises to strengthen the core muscles.
b) List one individual sport and one team sport.

2. Health & Hygiene
a) Why is regular exercise important for teenagers?
b) Mention two ways to prevent skin infections.

3. Safety & Environment
a) State one safety precaution while cycling outdoors.